

Wentao Shang

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 Wentao Shang |  DonaldShawnShang

Beijing / Shenzhen / Guangzhou - China

EDUCATION

- **Sun Yat-sen University** Sep 2024-Present
Eng.D in Computer Technology Guangzhou, China
 - GPA: 4.05/4.3
- **Pengcheng Laboratory** Sep 2025-Present
Associate Eng.D with Pengcheng Laboratory Shenzhen, China
 - RESEARCH ASSISTANT IN DEPARTMENT OF NEW NETWORKS
- **Beijing Jiaotong University** Sep 2020 - Jun 2024
BEng in Information Security (Secret Protection) Beijing, China
 - GPA: 3.49/4.00

RESEARCH EXPERIENCE

- **Unified Cross-Layer Intrusion Detection for Automotive Ethernet** Sep 2024 - Present
Tools: Transformer, Byte-level Tokenization, Knowledge Distillation, SOME/IP, TSN, Python, PyTorch
 - Proposed AutoGuard, a unified multi-protocol IDS for Automotive Ethernet using a byte-level Transformer with hex-bigram tokenization, achieving competitive detection across eleven attack types.
 - Compressed the model from 132M to 1.5M parameters via teacher-student distillation ($\sim 90\times$ reduction) while retaining competitive accuracy for deployment on resource-constrained zonal controllers.
 - **Published in *Computer Networks* (Elsevier, CCF-B).**

PUBLICATIONS AND PATENTS

J=JOURNAL, P=PATENT

- [J.1] Wentao Shang, Shuyuan Jin, Weihong Han. **AutoGuard: Unified and Lightweight Cross-Layer Intrusion Detection for Automotive Ethernet via Contextual Traffic Analysis**. *Computer Networks* (Elsevier, CCF-B), Vol. 286, Article 112394, 2026. DOI: 10.1016/j.comnet.2026.112394
- [P.2] Shuyuan Jin, Wentao Shang(2025-08-22). **A CAN bus Intrusion Detection Method based on Adaptive Unscented Kalman Filter (AUKF)**
- [P.1] Zhaoxi Li, Wentao Shang (2023-03-24). **Network Switch**. Patent Office: China National Intellectual Property Administration, Patent No. ZL 2022 2 3126982.8. Registration Date: 2022-11-24, Grant Date: 2023-03-24, Publication Date: 2023-03-24.

HONORS AND AWARDS

- **Third Prize in the 16th National Cybersecurity Competition** 2023
Awarded by Zhejiang University
- **Global Runner-up in BCTFxDEF CON 30 Auto-driving Contest** 2022
Awarded by Baidu-Security(BCTF) x UCI
- **BJTU Elite Award (Top 6 Students in the University)** 2023
Selected by Beijing Jiaotong University

TECHNICAL SKILLS

- **Programming:** Python, Shell, Git, Linux, Docker, OpenSandbox
- **ML & Data:** PyTorch, Hugging Face Transformers, PEFT (adapter fine-tuning), NumPy
- **LLM Security:** Prompt/Skill Injection, Guardrail Evaluation, Red Teaming, Safety Benchmarks

ACADEMIC SERVICE

JOURNAL REVIEWER:

- **Chinese Journal of Electronics** 2024
- **Neural Networks** 2026